

Pabil Adhikari

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EDUCATION

University of Colorado Boulder

MS. Major in Computer Science, GPA: 4.0/4.0

Boulder, CO

May 2026 - Present

Simon Fraser University

BSc. Major in Biochemistry and Certificate in Computing Science

Burnaby, BC

Sept 2019 - Sept 2025

PROJECTS (VIEW MORE AT PABILADHIKARI.DEV)

Running Analytics API | TypeScript, AWS (EC2, RDS), Express, PostgreSQL, Knex, React

- Built a **full-stack** analytics platform that authenticates with the **Strava API** via **OAuth2** and syncs personal training data into a **PostgreSQL** database hosted on **AWS RDS**, storing tokens for persistent state.
- Designed and wrote **raw SQL aggregation queries** for weekly mileage, pace trends, and personal record detection, exposed through a **TypeScript/Express REST API** deployed on an **AWS EC2** vm instance.
- Built a **React/TypeScript** frontend using functional components and hooks to consume the live API and visualize weekly mileage, pace, and personal record trends, completing an end to end **cloud deployed** full stack pipeline.

Discoverify | Python, Flask, Node.js, Spotify API, Render, REST APIs, Microservices

- Built a **full-stack** music recommendation platform with **Spotify OAuth** authentication and a **RESTful API** layer, enabling personalized playlist generation based on a user's real time listening history and genre preferences.
- Trained a **collaborative filtering model (SVD + KNN)** on **15,000 playlists** from Spotify's Million Playlist Dataset, systematically tuning hyperparameters via cross validation to optimize recommendation accuracy.
- Deployed a **microservices architecture** on Render using **agile** development practices, separating Python ML API and Node.js services and resolving cross service authentication challenges for a **production ready deployment**.

Protein Visualizer | Python, PyMOL, Requests API, Git

- Identified a recurring pain point in coursework by having to manually navigate the **RCSB PDB database** to search for, download, and load individual protein structure files, and built a **command line tool** to automate the entire pipeline from protein ID input to **3D PyMOL rendering** in a single command.
- Integrated the **Requests** library to programmatically fetch and parse **PDB files** directly from the RCSB API, automating retrieval and local file management to eliminate manual downloads entirely.
- Shared the tool with classmates, resulting in organic adoption by **40+ independent users** with no promotion.

WORK EXPERIENCE

Undergraduate Researcher

Gries Lab, Simon Fraser University

Burnaby, BC

Apr 2021 - Jan 2022

- Led a team of research assistants in the **placement, management and deployment** of field equipment, stepping into a **team lead role for 4 months** during the Principal Investigator's absence.
- Analyzed experimental data to refine strategies, resulting in a **30% improvement in operational efficiency** and a **60% increase in capture rates**, significantly enhancing research accuracy and reproducibility.
- Mentored new research assistants on experimental methodology and communicated experimental objectives and outcomes across **senior and junior lab management**, demonstrating **cross functional collaboration**.

PUBLICATION

Title: 1-Hexanol and 2,3,5-trithiahexane - novel sex attractant 2 pheromone components of wild house mouse males.

Authors: Elana K. Varner, Daniella Gofredo, Kaya Vukovic, Kendal C. Y. Singleton, **Pabil Adhikari**, Et al.

TECHNICAL SKILLS

Computer Languages: Python, JavaScript, TypeScript, C, SQL, HTML, CSS

Frameworks & Libraries: React, Node.js, Express, Flask, Knex, PyTorch, NumPy

Developer Tools: Git, GitHub Actions, REST APIs, Linux, pm2

Cloud & Infrastructure: AWS (EC2, RDS), GCP, Terraform